

ResStock-LLM: A Multi-Agent Framework for Climate-Adaptive Residential Retrofit Decisions

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Abstract. Residential building retrofits are essential for improving energy efficiency and reducing greenhouse gas emissions, yet identifying effective retrofit actions for building stocks remains challenging. Current methods often compare pre- and post-retrofit simulations, thereby ignoring regional differences and the distribution of efficiency gaps. To improve automation in retrofit decision-making, this study introduces ResStock-LLM, a large language model (LLM) framework that integrates multiple agents with ResStock data to analyze household descriptions, forecast building energy-efficiency percentiles, and develop climate-specific retrofit strategies. ResStock-LLM links pre-trained building energy-efficiency classifiers to assess retrofit needs and compares them against national and zone-level building-stock data from ResStock. The identified retrofit targets are directed to a report agent that generates code-compliant retrofit reports. Using the structural knowledge base and ResStock data, this approach demonstrates that LLM agents can produce reliable recommendations. Tests on a representative single-family residential building show that ResStock-LLM primarily identifies roof insulation, heating setpoint, and shading improvements as key factors influencing retrofit potential. In general, ResStock-LLM offers a scalable, climate-adaptive decision-support tool that integrates building stock models with a pre-trained machine learning classifier and language model-based reasoning to facilitate efficient retrofit planning.

Keywords. Building retrofit; Large language model (LLM); Building energy efficiency; agentic artificial intelligence; Automatic decision-making.

1 Introduction

Retrofitting residential buildings is a practical way to reduce global energy use and carbon emissions [1]. However, the current decision-making framework for retrofitting comprises multiple, separate stages, from on-site assessment and energy audits to retrofit analysis and planning. The fragmented framework requires substantial time and human intervention, making residential retrofit impractical and expensive for most households. This prolongs the service life of outdated construction that remains underperforming. Automatic, adaptive frameworks for retrofitting are critical to addressing this challenge and accelerating progress toward building sustainability and low-carbon emissions. However, it remains challenging to systematically determine building retrofit actions at a large scale, given the diversity of the building stock, family demographics, and regional climates.

Traditional assessment approaches involve comparing pre- and post-retrofit energy use by performing physics-based building energy simulations (BES) for individual buildings [2]. While BES can compare energy savings, it requires specialized expertise and intensive computational

resources. This essentially limits the generalizability of BES in large-scale building retrofits. Besides, current retrofit approaches primarily focus on energy reduction while ignoring broader factors that influence residential retrofit decision-making in the real world. For example, different energy-efficiency benchmarks yield different decisions about building retrofits. Moreover, households with varying incomes have different preferences for home retrofitting.

Large language models (LLMs) have demonstrated remarkable capabilities in multitasking and automation across scientific fields, from computer science to biomedical analysis. LLM-assisted decision-making can bridge the gap between data-driven prediction and human-like reasoning, enabling scalable, tailored automation in specific contexts. These advances in agentic artificial intelligence (AI) open new opportunities for context-aware decision systems in the built environment [3].

This research proposes an LLM-integrated decision-support framework, ResStock-LLM, which combines structured feature extraction, data-driven prediction, and SHAP-guided reasoning for interactive retrofit assessment. ResStock-LLM conducts scalable, interpretable, and user-interactive retrofit pre-screening

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without requiring full simulation modeling, making it particularly suitable for early-stage assessments, policy screening, and homeowner-level decision support. Specifically, ResStock-LLM uses national-scale energy and housing data to predict the inefficient probability of an individual home relative to climatic benchmarks and reason about optimal retrofit strategies informed by regional priorities and data-driven feature diagnostics (i.e., SHAP analysis). By coupling data-driven classification with language-based reasoning, ResStock-LLM can provide climate-specific, explainable, and adaptable retrofit recommendations at the individual-building level.

The remainder of this paper is organized as follows: Section 2 reviews related work on building retrofit strategies. Section 3 outlines the ResStock-LLM workflow and details the role of each agent in the proposed architecture. Section 4 presents a case study that demonstrates the performance and reasoning capabilities of ResStock-LLM across representative U.S. climate zones. Section 5 concludes with findings, limitations, and future opportunities in agentic-based decision-making.

2 Related Work

2.1 Audit-based retrofit approaches

Residential energy retrofits have largely been guided by on-site audits and expert heuristics. Professional on-site energy audits serve as the first step in identifying retrofit targets and recommending upgrade options, which rely on rules-of-thumb and checklists. Common upgrades include insulation improvements, HVAC tune-ups, and window replacements based on observed deficiencies and payback estimates [4, 5]. Such audit-based methods generally require domain expertise and standardized guidelines, which are limited in scope. Retrofit plans are often heuristic, focusing on the most evident drivers of inefficiency. Although audit-based methods are practical, they face challenges in consistency and optimality. Solutions applicable to one residence or climate condition may not be suitable for others. Moreover, such audit data can be erroneous and subjective. Conventional energy-efficiency assessment has poor generalizability and interpretability, underscoring the need for more systematic retrofit design methods.

2.2 Data-driven retrofit approaches

Over recent years, the field of building retrofit has seen rapid expansion in data-driven, large-scale modeling approaches. Building retrofit research has been addressing the complexity, nonlinearity, and scale of retrofit decision-making. A comprehensive review by Bocaneala et al. [6] synthesizes over decades of AI-based retrofit studies and identifies dominant application algorithms, including machine learning, intelligent optimization, surrogate modeling, fuzzy logic, and knowledge discovery, as well as building automation systems and

knowledge-based systems. Across these algorithms, AI has primarily been used to improve prediction accuracy and reduce computational cost in an uncertain manner [7, 8].

A broader range of real-world variability factors has been identified as influential and incorporated into retrofit building planning, such as occupant behavior and technology adoption. For example, Chen et al. [9] introduced stochastic occupant behavior models into a bottom-up stock simulation to capture how different usage patterns affect energy savings. Climate change is significant to building energy performance. Luo and Oyedele [10] considered the climate change effects to optimize the life cycle for building retrofit. Economic feasibility is a key determinant of retrofit adoption. Integrating these economic models with energy performance assessments enables the generation of retrofit plans that include initial cost estimates and return on investment metrics, supporting informed decision-making. Hong et al. [11] conducted lifecycle costing analysis using EnergyPlus simulation outputs, investment cost, and operational data, aiming to evaluate retrofit payback periods and carbon reduction. Moreover, the Lawrence Berkeley National Laboratory (LBNL) [12] provided authoritative U.S.-focused retrofit cost data and decarbonization scenarios, which are essential for regional adaptation. Accurate household profiling is foundational for personalized retrofit recommendations. To complement this, Ma et al. [13] provided a comprehensive review of building floorspace and stock measurements, emphasizing the importance of accurate building characterization for retrofit planning. These studies highlight the feasibility of constructing detailed household profiles by combining building physical attributes with occupant data, supported by synthetic data augmentation to address data scarcity.

2.3 LLM-integrated decision support

LLMs have recently demonstrated their ability to multitask and learn from text corpora. This generalist training enables LLMs to perform cross-domain reasoning: they can integrate knowledge about materials, climate, economics, and user preferences into a single coherent response. With appropriate prompting, they perform step-by-step logical deductions and quantitative reasoning, akin to an expert working through a problem. Additionally, LLMs can analyze real-world data and output human-readable explanations. In other words, an LLM can translate the outcomes of engineering analyses into plain-language recommendations. Their multitasking and explanatory capabilities have already shown promise in high-stakes fields: for instance, an advanced LLM with enhanced reasoning matched human experts in a clinical decision-support setting. In the building domain, researchers envision LLMs as conversational partners for planners and homeowners, justifying appropriate retrofit measures at both ends. Crucially, LLMs can incorporate

comprehensive criteria (such as energy, cost, comfort, and climate) and provide a narrative rationale, helping decision-makers understand why a particular retrofit measure is suggested.

2.4 Research gap

Although LLM applications in residential retrofit planning are emerging, there remains a significant gap in providing generalizable, climate-adaptive tools. Most prior studies treat homes as a whole and offer heuristic recommendations. There has been little exploration of tailored LLMs, for example, to differentiate among building envelope upgrades, HVAC system improvements, and window replacements. Addressing this gap is vital to developing customized retrofit plans that account for subsystem priorities, climate zones, and building types, enabling more effective, context-aware advice. Translating LLMs into specialized planning tools remains a key opportunity for innovation in AI and building science.

3 Methodology

ResStock-LLM is a multi-agent decision-support tool that integrates pre-trained machine-learning classifiers and LLM agents to automate residential building retrofits. As shown in Figure 1, the system ingests unstructured building descriptions provided by homeowners, predicts building energy efficiency, and generates homeowner-

ready retrofit reports. The input-output protocol is managed by four agents:

- (1) Intake agent extracts household and home attributes from unstructured descriptions and stores the information as a schema-compliant JSON record.
- (2) Scoring agent takes the JSON record into a pre-trained classifier, which can determine the likelihood of residential energy inefficiency.
- (3) Reasoning agent adopts SHAP analysis to the pre-trained classifier, which provides feature importance on building inefficiency and interprets retrofit priorities.
- (4) Report agent generates homeowner-ready retrofit reports.

All components were orchestrated using Open-AI as a sequence of LLM tools and deterministic pipelines.

3.1 Household profiling

This intake agent is an LLM-based information-extraction module constrained by a strict schema. It interprets free-text descriptions such as “*We live in a 1985 two-story house in climate zone 5 with central AC*” and converts them into structured features for income, vintage, building geometry, equipment setpoints, construction types, and climate zone. If user descriptions omit required attributes, the agent either infers missing values using LLM-based reasoning (e.g., deriving climate zone from ZIP code) or flags the input as invalid and prompts the user for clarification. Only validated, schema-consistent JSON records are passed downstream.

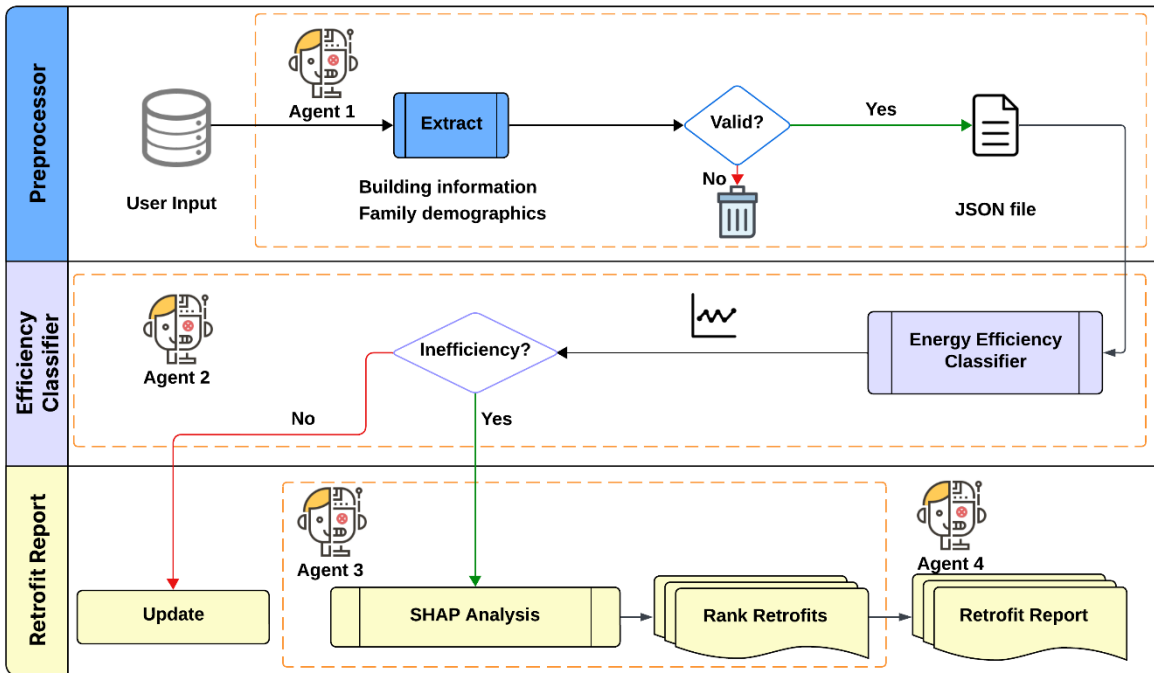


Fig. 1. ResStock-LLM decision support pipeline in three sequential modules. Each module is monitored by an agent that parses inputs, classifies efficiency, diagnoses building feature deficiencies, and generates an explanatory report.

3.2 Energy efficiency assessment and retrofit necessity

The scoring agent loads an encapsulated end-to-end ML pipeline to evaluate whether the profiled home is likely to be energy-inefficient and therefore a candidate for retrofit. Specifically, the scoring agent encodes the JSON output from the intake agent, calculates the probability of inefficiency, and returns a binary classification based on pre-defined climate-specific decision thresholds. The pre-trained XGB classifier was trained on approximately 150,000 homes from the ResStock database and achieves 87% accuracy in predicting inefficient homes across U.S. climates. It can provide robust, regionally relevant assessments across climatic contexts.

3.3 Feature-level retrofit target identification

The reasoning agent uses SHAP (SHapley Additive exPlanations) to translate classifier predictions into feature-level retrofit targets. For each individual building, local SHAP values are computed to decompose the classifier's predicted inefficiency probability into additive feature contributions relative to a background data distribution. SHAP quantifies how each input feature shifts the model's predictions away from its expected baseline. Positive SHAP values indicate features that increase the predicted probability of inefficiency, while negative values indicate features that mitigate it. Based on the magnitude of SHAP values, the agent ranks retrofit priorities by identifying the features that most strongly drive the inefficiency prediction for that specific dwelling. This enables ResStock-LLM to produce instance-specific retrofit recommendations grounded in the building's actual configuration.

3.4 Homeowner-ready retrofit report generation

The report agent accesses the latest construction code and extracts the retrofit standards for identified retrofit targets. It then compiles all upstream outputs, including the household profile, inefficiency classification, SHAP-ranked priorities, and retrofit goals, into a comprehensive retrofit report tailored for homeowners. The report will be organized as follows: (1) an executive summary outlining the retrofit need; (2) an accessible explanation of the inefficiency risk and key driving factors; (3) a ranked list of recommended retrofit actions; and (4) climate-zone and construction-compliant references.

This agent interprets abstract ML outputs into clear, actionable guidance to support homeowners' decision-making and align with regional best practices for residential energy upgrades.

4 Case study

This case study illustrates the internal reasoning workflow of ResStock-LLM through a representative residential

building. The objective here is to demonstrate how a multi-agent LLM streamlines the process of extracting building information, identifying dominant inefficiency drivers, and generating retrofit recommendations, rather than evaluating energy savings. The selected building represents a single-family detached home located in IECC Climate Zone 5, randomly selected from the ResStock database. The building was constructed prior to 1990 and exhibits characteristics commonly associated with inefficient residential stock in cold climates. Key envelope and system features for inference and reasoning include wall insulation, roof insulation, inefficient windows, air infiltration, and HVAC system efficiency.

4.1 Intake agent

Given a free-text household description (see Appendix), the intake agent extracted key information and generated a structured summary describing thermal, operational, and household conditions. The extracted features were stored in a structured JSON file and passed to the scoring agent.

4.2 Scoring agent

The scoring agent called the pre-trained ML model to assess the residential energy inefficiency based on the extracted building summary. The inefficiency probability of this building is 33.57%, which exceeds the pre-defined threshold (19.17%) in climate zone 5. Therefore, this building was labelled as inefficient, which is considered a retrofit candidate.

4.3 Reasoning Agent

The reasoning agent calculated SHAP values to analyze feature contributions to inefficiency. These values were derived as local additive explanations for each input, estimating feature impacts by marginalizing over the background data distribution. SHAP analysis revealed a set of building components associated with energy inefficiency: roof construction type, heating setpoint, and building orientation. Roof construction requires the highest-leverage intervention because its insulation directly affects solar-gain control. The heating setpoint was considered the second priority, indicating that heating thermal comfort plays a significant role in this building's inefficiency. However, heating system replacement is recommended only conditionally, contingent on envelope improvements that reduce load requirements. Building orientation largely increased the risk of inefficiency; therefore, shading strategies and solar-gain control can improve thermal resistance and reduce base-heating demand.

4.4 Report Agent

The report agent extracted useful information and generated a concise, human-readable recommendation on

building retrofit. For this case, the recommended retrofit sequence is:

1. Roof construction type: Increase attic/roof insulation to the recommended R-value for CZ 5.
2. Heating setpoint: Lower heating setpoint during unoccupied periods or at night, and install a programmable or smart thermostat with setbacks.
3. Orientation: Add exterior shading devices on high-gain façades and use low-SHGC glazing or shading films on east/west windows.

Each recommendation included a brief explanation that connects the action to climate-specific drivers of heating demand and inefficiency. No quantitative energy savings are reported, as the case study emphasizes interpretability over performance estimation.

Table 1. Summary of Agent Outputs for the Representative Building Case Study

Agent	Input	Output	Interpretation
Intake agent	Building information	Building condition summary	Identifies insufficient wall insulation and high infiltration as dominant weaknesses for a pre-1990 home in Climate Zone 5.
Scoring agent	Building features	Energy efficiency state and primary drivers of inefficiency	Roof heat loss and heating setpoint are classified as high impact; building orientation effects are secondary.
Reasoning agent	Climate-adaptive drivers of inefficiency	Retrofit ranking	Determines that roof upgrades are prerequisites for effective envelope improvements; avoids isolated HVAC retrofit.
Report agent	Retrofit decision logic	Final retrofit report	Prioritizes roof insulation, thermostat installation, and shading strategies, with heating system upgrade conditional on envelope improvements.

5 Conclusion

This paper introduces ResStock-LLM, a data-driven, interpretable framework for climate-specific residential retrofit reasoning. It combines large-scale building data, distributional analysis, and LLM reasoning to identify inefficiency patterns and prioritize retrofits without building-by-building optimization or high-fidelity simulations. A case study in climate zone 5 shows that ResStock-LLM can generate climate-aware retrofit recommendations that reflect the relative importance of key features. The results demonstrate the potential of LLMs as reasoning layers in building energy decision-support systems, complementing traditional methods. Future work will expand to more climate zones, include cost and equity factors, and integrate LLM reasoning with physics-based simulation to support scalable, policy-relevant retrofit planning.

Acknowledgment

This work was supported by the National Science Foundation under Grant No. 2515145 (Collaborative Research: CAS: Examining carbon emission implications of the impact of extreme weather conditions on building material durability, aging, and deterioration).

References

4.5 Discussion

This case study illustrates that ResStock-LLM can enable interpretable, climate-aware retrofit reasoning by linking statistical feature disparities to domain-relevant retrofit actions, as shown in Table 1. ResStock-LLM does not require calibrated energy models to simulate energy savings before and after retrofit actions across a large group of residential buildings. Instead, it leverages existing large-scale data to identify the dominant drivers of inefficiency and uses LLM-based reasoning to synthesize these insights into actionable guidance. While the study is limited to one climate zone and a subset of building features, it demonstrates the scalability and transparency of agentic AI in automated retrofit decision-making.

1. Z. Li, Y. Han, P. Xu, Methods for benchmarking building energy consumption against its past or intended performance: An overview. *Appl. Energy*. 124, 325–334 (2014). <https://doi.org/10.1016/J.APENERGY.2014.03.020>.
2. E. Lo Cascio, Z. Ma, D. Borelli, C. Schenone, Residential Building Retrofit through Numerical Simulation: A Case Study. *Energy Procedia*. 111, 91–100 (2017). <https://doi.org/10.1016/J.EGYPRO.2017.03.011>.
3. L. Zhang, V. Ford, Z. Chen, J. Chen, Automatic building energy model development and debugging using large language models agentic workflow. *Energy Build.* 327, 115116 (2025). <https://doi.org/10.1016/j.enbuild.2024.115116>.
4. C. Chen, J. Wang, H. Zhang, X. Xu, L. E. Hinkle, X. Chao, Q. Shi, Dual impacts of solar-reflective façades in high-density urban areas on building energy use and outdoor thermal environments. *Energy Build.* 324, 114926 (2024). <https://doi.org/10.1016/j.enbuild.2024.114926>.
5. X. Xu, J. Wang, Comparative Analysis of Physics-Guided Bayesian Neural Networks for Uncertainty Quantification in Dynamic Systems. *Forecasting*. 7, 9 (2025). <https://doi.org/10.3390/forecast7010009>.
6. N. Bocaneala, M. Mayouf, E. Vakaj, M. Shelbourn, Artificial Intelligence Based Methods

- for Retrofit Projects: A Review of Applications and Impacts. Archives of Computational Methods in Engineering. 32, 899–926 (2025). <https://doi.org/10.1007/s11831-024-10159-7>.
7. X. Xu, Y. Hu, S. Atamturktur, L. Chen, J. Wang, Systematic review on uncertainty quantification in machine learning-based building energy modeling. Renewable and Sustainable Energy Reviews. 218, 115817 (2025). <https://doi.org/10.1016/j.rser.2025.115817>.
 8. X. Xu, J. Wang, Information-theoretic and Bayesian model selection for physics-based modeling: Balancing fit, complexity, and generalization. Inf. Sci. (N. Y). 726, 122743 (2026). <https://doi.org/10.1016/J.INS.2025.122743>.
 9. J. Chen, R. Adhikari, E. Wilson, J. Robertson, A. Fontanini, B. Polly, O. Olawale, Stochastic simulation of occupant-driven energy use in a bottom-up residential building stock model. Appl. Energy. 325, 119890 (2022). <https://doi.org/10.1016/j.apenergy.2022.119890>.
 10. X. J. Luo, L. O. Oyedele, Life cycle optimisation of building retrofitting considering climate change effects. Energy Build. 258, 111830 (2022). <https://doi.org/10.1016/J.ENBUILD.2022.111830>.
 11. Y. Hong, C. I. Ezech, W. Deng, S. H. Hong, Y. Ma, Y. Tang, Y. Jin, Coordinated energy-environmental-economic optimisation of building retrofits for optimal energy performance on a macro-scale: A life-cycle cost-based evaluation. Energy Convers. Manag. 243, 114327 (2021). <https://doi.org/10.1016/J.ENCONMAN.2021.114327>.
 12. S. Baik, K. Chelminski, S. Earni, E. Stuart, Evaluating GHG Mitigation Potential from ESPC Projects.
 13. M. Ma, S. Zhang, J. Liu, R. Yan, W. Cai, N. Zhou, J. Yan, Building floorspace and stock measurement: A review of global efforts, knowledge gaps, and research priorities. . 2, 100075 (2025). <https://doi.org/10.1016/J.YNEXS.2025.100075>.

Appendix

Intake agent input:

"We live in a two-story detached home built around 1985 in the State College, PA area, ZIP 16803. The house is about 2,200 square feet with 3 bedrooms and a mostly finished basement. The foundation is poured concrete. The walls are 2x4 wood framing with older fiberglass insulation, and the roof uses asphalt shingles. Our household income is approximately \$95,000. We typically heat the house to 74°F during winter and cool it to around 72°F in summer. The home faces southwest."

Intake agent output:

JSON file: {'valid': True, 'missing': [], 'features': {'Income': '60000-99999', 'Vintage': '1981-2000', 'Bedroom count': 3, 'Story count': 2, 'Heating setpoint': 74, 'Cooling setpoint': 72, 'Climate zone': 'CZ5', 'Foundation construction type': 'Basement', 'Orientation': 'Southwest', 'Roof construction type': 'Asphalt', 'Wall construction type': 'Wood Frame'}}

Scoring agent output:

Pre-trained ML model output: {'retrofit_probability': 0.3357487618923187, 'threshold': 0.1916583925485611, 'zone_used': 'CZ5', 'label': 1, 'label_meaning': '1 = inefficient (retrofit recommended), 0 = efficient'}

Report agent output:

SHAP: {'summary': 'Building classified as inefficient; the following actions are prioritized based on SHAP importance.', 'retrofit_probability': 0.3357487618923187, 'zone': 'CZ5', 'threshold': 0.1916583925485611, 'label': 1, 'label_meaning': '1 = inefficient (retrofit recommended); 0 = efficient', 'ranked_actions': [{'feature': 'Roof construction type', 'current_value': 'Asphalt', 'shap_value': 0.1837833821773529, 'description': 'Roof/attic insulation and solar gain control', 'recommended_actions': ['Increase attic/roof insulation to recommended R-value for the zone.', 'Consider cool roof materials or reflective coatings if cooling-dominated.']}, {'feature': 'Heating setpoint', 'current_value': 74, 'shap_value': 0.05421227216720581, 'description': 'Heating setpoint and thermostat control', 'recommended_actions': ['Lower heating setpoint during unoccupied periods or at night.', 'Install a programmable or smart thermostat with setbacks.']}, {'feature': 'Orientation', 'current_value': 'Southwest', 'shap_value': 0.04123637080192566, 'description': 'Solar gains and shading strategies', 'recommended_actions': ['Add exterior shading devices on high-gain façades.', 'Use low-SHGC glazing or shading films on east/west windows.']}]}

LLM-generated text report

Residential Energy Retrofit Report

Executive Summary

This report provides a tailored set of energy retrofit recommendations for your home, based on a data-driven assessment of its current features and energy performance. Using advanced modeling and SHAP-based analysis, we have identified the most impactful areas for improvement. The recommendations are prioritized to maximize energy savings, comfort, and cost-effectiveness, with specific guidance for your climate zone (CZ5) and home characteristics.

Building Profile

- **Location Climate Zone:** CZ5 (Cold/Moderate)
- **Home Vintage:** 1981–2000
- **Bedrooms:** 3
- **Stories:** 2
- **Foundation Type:** Basement
- **Wall Construction:** Wood Frame
- **Roof Construction:** Asphalt
- **Orientation:** Southwest
- **Thermostat Setpoints:** Heating 74°F, Cooling 72°F
- **Household Income:** \$60,000–\$99,999

Model Assessment Overview

A predictive model evaluated your home's energy efficiency, considering construction details, thermostat settings, and orientation. The model assigned a retrofit probability of 0.34, which is above the threshold of 0.19, indicating that your home is likely inefficient and would benefit from targeted energy upgrades. The following sections detail the most influential factors and recommended actions.

SHAP-Guided Retrofit Priorities

Priority 1: Roof Construction Type

- **Current Condition:** Asphalt roof
- **SHAP Value:** Strong driver of inefficiency
- **Why It Matters:** The roof and attic are major pathways for heat loss in winter and heat gain in summer, especially in cold climates like CZ5. Asphalt roofs can absorb significant solar heat, increasing cooling loads, and may lack sufficient insulation.

- **Recommendations:**

1. **Increase attic/roof insulation** to the recommended level for CZ5. This is a **capital upgrade** that can significantly reduce heating and cooling costs.

2. **Consider cool roof materials or reflective coatings** if your home experiences high summer cooling loads. This is a **capital upgrade** that helps reflect solar radiation and reduce attic temperatures.

Priority 2: Heating Setpoint

- **Current Condition:** 74°F
- **SHAP Value:** Moderate driver of inefficiency
- **Why It Matters:** Higher heating setpoints increase energy use, especially during cold months. Maintaining a high temperature throughout the day and night can lead to unnecessary heating costs.

- **Recommendations:**

1. **Lower the heating setpoint during unoccupied periods or at night.** This is a **behavioral measure** that can yield immediate savings.

2. **Install a programmable or smart thermostat with setback features.** This is a **low-cost upgrade** that automates temperature adjustments for efficiency.

Priority 3: Orientation

- **Current Condition:** Southwest-facing
- **SHAP Value:** Small but notable driver of inefficiency
- **Why It Matters:** Southwest orientation can lead to increased solar heat gain, particularly in the afternoons, affecting both heating and cooling loads. Managing sunlight through windows is important for comfort and efficiency.

- **Recommendations:**

1. **Add exterior shading devices** (such as awnings or shutters) on high-gain façades. This is a **capital upgrade** that reduces unwanted solar gain.

2. **Use low solar heat gain coefficient (SHGC) glazing or shading films** on east and west windows. This is a **capital or low-cost upgrade** depending on the chosen solution.

Climate-Zone-Aware Guidance

Your home is located in Climate Zone 5, which experiences cold winters and moderate summers. For this zone:

- **Roof/Attic Insulation:** Upgrading insulation is especially important to prevent heat loss during long heating seasons.

- **Cool Roofs:** While more critical in hot climates, reflective roofing can still benefit homes with high summer cooling needs.

- **Thermostat Management:** Heating dominates energy use; lowering setpoints and using setbacks are highly effective.

- **Solar Gain Control:** Southwest orientation requires attention to window shading and glazing to balance winter heat gain and summer overheating.

Recommended Implementation Roadmap

1. Immediate Actions (Low-cost/Behavioral):

- Lower heating setpoint during unoccupied periods or at night.

- Begin using existing thermostat setback features if available.

2. Short-Term Upgrades (Low-cost/Capital):

- Install a programmable or smart thermostat.

- Apply window films or install interior shades on southwest-facing windows.

3. Medium-Term Upgrades (Capital):

- Increase attic/roof insulation to meet or exceed CZ5 recommendations.

- Add exterior shading devices to high-gain façades.

4. Long-Term Considerations:

- When replacing the roof, consider cool roof materials or reflective coatings to further reduce cooling loads.

Notes & Limitations

- This report is based on available data and model-driven analysis. Actual savings and comfort improvements may vary depending on installation quality, occupant behavior, and other factors.

- Recommendations are aligned with best practices for your climate zone but do not substitute for a detailed on-site energy audit.

- Cost estimates and payback periods are not included; consult local contractors or energy advisors for project-specific details.

- Some measures may be eligible for local incentives or rebates—check with your utility or state energy office.

For further guidance or to begin implementing these recommendations, consider contacting a certified energy auditor or retrofit contractor familiar with homes in your region.